MEDINOVA TEACHING HOSPITAL

Class 02 — Investigations & Interpretation

Laboratory Results · Imaging · Clinical Reasoning

> *"The investigation serves the history, never the reverse."*

CLERKSHIP SERIES | SESSION ID: 400-INV-02

SECTION 01

The Diagnostic Hierarchy

In the modern clinical setting, there is a temptation to "screen" patients with a battery of tests before a physical examination. This is **diagnostic noise**. The hierarchy of importance remains:

- **History (80%):** Defines the pre-test probability.
- **Physical Exam (15%):** Refines the differential diagnosis.
- **Investigations (5%):** Confirms or excludes specific hypotheses.

An investigation should only be ordered if its result has the potential to change your management plan.

SECTION 02

Full Blood Count (FBC) Interpretation

Parameter	Clinical Significance of Abnormality	Medical Meaning
Hb (Haemoglobin)	Low: Anaemia (blood loss, deficiency, chronic disease). High: Polycythemia.	The protein in red cells that carries oxygen.
WBC (White Cells)	High: Infection, inflammation, or Leukaemia. Low: Bone marrow failure.	The immune system cells that fight "invaders."
MCV (Mean Cell Vol)	Low: Microcytic (Iron def). High: Macrocytic (B12/Folate def, Alcohol).	The average size of a single red blood cell.
Platelets	Low: Thrombocytopenia (risk of bleeding). High: Inflammation/Malignancy.	Tiny cell fragments that help the blood clot.
Neutrophils	High: Specifically suggests acute bacterial infection.	A specific type of white cell first on the scene of infection.

PROF. OBI'S PEARL #1

"Never treat the machine's printout. If the Hb is 6.0 but the patient is jogging in the corridor, check the name on the tube before you order a transfusion."

SECTION 03

Arterial Blood Gas (ABG): The 4-Step Approach

An ABG measures the acid-base balance and gas exchange in the blood. Follow these steps strictly:

1. **pH:** Is it Acidosis (< 7.35) or Alkalosis (> 7.45)?
2. **$PaCO_2$:** Does the CO_2 explain the pH? (Respiratory)
3. **HCO_3:** Does the Bicarbonate explain the pH? (Metabolic)
4. **Compensation:** Is the other system trying to fix the mess?

The 6 Primary Disorders

Disorder	pH	Primary Driver	Common Example
Resp. Acidosis	Low	High CO_2	COPD, Opioid Overdose
Resp. Alkalosis	High	Low CO_2	Panic Attack (Hyperventilation)
Met. Acidosis	Low	Low HCO_3	Diabetic Ketoacidosis (DKA)
Met. Alkalosis	High	High HCO_3	Severe Vomiting (loss of stomach acid)
Mixed Disorder	Var.	Dual drivers	Salicylate (Aspirin) poisoning
Compensated	Normal	Balanced	Chronic lung disease

Acidosis: Too much acid in the blood.
Alkalosis: Too much base/alkali in the blood.
$PaCO_2$: Partial pressure of Carbon Dioxide (regulated by lungs).
HCO_3: Bicarbonate (regulated by kidneys).

SECTION 04

Chest X-Ray: Systematic ABCDE Review

A structured approach ensures you never miss subtle findings. *Always verify patient name, date, and film quality (R.I.P: Rotation, Inspiration, Penetration) first.*

System	What to look for	Clinical Hint
A - Airway	Tracheal position, Carina, Bronchi.	Is the trachea shifted? (Suggests Tension Pneumothorax).
B - Breathing	Lung fields, Pleura, Hila.	Look for "White-outs" (Pneumonia) or "Black-outs" (Pneumothorax).
C - Cardiac	Heart size and shape.	Cardiomegaly (Enlarged heart) if diameter > 50% of chest.
D - Diaphragm	Costophrenic angles, Gastric bubble.	Blunted angles suggest Effusion (fluid around lungs).
E - Everything Else	Bones, Soft tissue, Medical lines.	Rib fractures or pacemaker leads.

PROF. OBI'S PEARL #2

"A Chest X-ray is a 2D representation of a 3D patient. If you're unsure of a shadow, go back to the bedside and listen for a crackle. The stethoscope often sees better than the film."

SECTION 05

Renal Function & Electrolytes (U&E)

Monitoring the "Chemistry" of the blood is vital for drug dosing and acute care.

- **Creatinine:** The waste product of muscle. Best indicator of kidney filter health.
- **Urea:** Elevated in dehydration or GI bleeds.
- **eGFR:** Estimated Glomerular Filtration Rate. (Normal > 90. Kidney failure < 15).
- **Potassium (K+):** The most dangerous electrolyte. High K+ (Hyperkalaemia) can stop the heart. Threshold for emergency: > 6.0 mmol/L.

SECTION 06

Imaging Modalities: When to Request What

Modality	Indications	Radiation Risk
X-Ray	Bones, Chest, Bowel gas.	Low (equivalent to 3 days background radiation).
CT Scan	Trauma, Brain bleeds, Complex Abdomen.	High (equivalent to ~200 X-rays). Use sparingly.
MRI	Spinal cord, Soft tissue, Ligaments.	None (uses magnetic fields).
Ultrasound	Gallstones, Pregnancy, DVT.	None (uses sound waves).

PROF. OBI'S PEARL #3

"CT scans are wonderful, but they are not 'truth machines'. A negative CT brain does not always rule out a clinical stroke in the first few hours."

SECTION 07

Clinical Reasoning: Pre-test Probability

We use **Bayes' Theorem** intuitively every day. Your belief in a diagnosis after a test depends heavily on how much you believed it *before* the test.

Likelihood Ratio (LR): A measure of how much a test result changes the probability of disease.
• LR+ of 10 = Strong evidence for the disease.
• LR− of 0.1 = Strong evidence against the disease.

SECTION 08

Key References

- **NICE Guidelines:** NG127 – Suspected Cancer Recognition and Referral.
- **RCR (Royal College of Radiologists):** iRefer Guidelines for imaging standards.
- **BSH (British Society for Haematology):** Guidelines on the investigation of Anaemia.

> **PROF. OBI'S PEARL #4**
> "The most expensive test is the one you didn't need to order. The second most expensive is the one you ordered but didn't know how to interpret."

OFFICIAL SIGN-OFF

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